INCH-POUND

A-A-3153A
July 30, 2002
SUPERSEDING
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April 15, 1998

COMMERCIAL ITEM DESCRIPTION

BIT, SCREWDRIVER, MALE SHANK, HEXAGONAL TIP

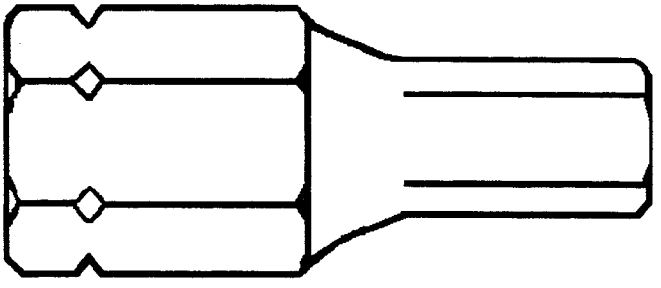
The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This Commercial Item Description covers hand driven male hexagonal point screwdriver insert bits.

2. CLASSIFICATION. Not applicable

3. SALIENT CHARACTERISTICS.

3.1 Design. Hexagonal tip insert bit with male hexagonal shank. Unless otherwise specified, all bits shall meet the requirements of ASME/ANSI B18.3 Socket Cap, Shoulder, and Set Screws (Inch Series) and ANSI B18.3.2M Metric Series Hexagon Keys and Bits as applicable. Bit dimensions shall conform to Table 1, Section 3.4.2.



Screwdriver Bit, Hexagonal Shank, Hexagonal Tip.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Hardware Superstore (6FEE), 1500 E. Bannister Rd. Bldg # 6, Kansas City, Missouri 64131.

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3.2 Material.

3.2.1 Inch Series. The material shall be in accordance with ASME/ANSI B18.3, Table 8, Note 3.

3.2.2 Metric Series. The material shall be in accordance with ANSI B18.3.2M, Paragraph 3.1, Material and Heat Treatment.

3.3. Construction.

3.3.1 Screwdriver Bits, Male Hexagonal. The screwdriver bits shall be hardened and tempered to comply with the specified physical and mechanical requirements. The cross-sectional shape of the blade shall be hexagonal. The bits shall be free from visible defects such as scale, seams, laps, and cracks.

3.3.1.1 Male Hexagonal Tip. The bit tip shall be of the type known commercially as a hex bit and be manufactured to install and remove hexagon head cap screws in accordance with ASME/ANSI B18.3.

3.3.1.2 Male Hexagonal Shank. The shank shall be hexagonal in shape with a retaining groove. Shape of the retaining groove is optional.

3.3.1.3 Shank With O-Ring. When specified, an O-ring retainer placed in a grove of like size, and a protruding positive stop, to accommodated a ratcheting box wrench will be provided.

3.3.1.4 Shank With Knurled or Serrated Spinner. When specified, bits shall be provided with a knurled or serrated spinner on the opposite end for use in a ratcheting box wrench. Bits shall be firmly and securely retained by a mechanical means incorporated in the hexagonal bit body. The overall length of bit and spinner shall not exceed 1-5/8 inches.

3.4 Dimensions.

3.4.1 Male Hexagonal Tip. The hexagon width across flats, hexagon width across corners, and chamfer dimension of the male hexagonal tip of the bit shall be in accordance with ASME/ANSI B18.3, Table 8, Dimensions of Hexagonal Keys and Bits (Inch) or ANSI B18.3.2M, Table 1, Dimensions of Metric Hexagonal Keys and Bits.

3.4.2 Bit length. The overall bit length shall conform to the dimensions in Table 1, or the overall bit length shall be as specified by the purchaser.

Table 1 Nominal Bit Length

Shank Size	**Nominal Bit Length**
Inches	Inches
1/4	1-1/8 +/- 3/16
5/16	1-1/4 +/- 3/16
7/16	1-3/8 +/- 1/4

3.5 Coating Requirements. The bits shall be coated with black oxide, black phosphate, or plain lightly oiled or coated with a rust preventative unless otherwise specified by the purchaser.

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3.6 Bit Hardness. The bits shall be through hardened 50 to 64 HRc scale taken 1/4 inch from the tip end unless otherwise specified by the purchaser. When tested in accordance with ASTM E-18, the hardness shall be measured as close to the tip as possible and shall be as specified.

3.7 Bit Torsional Requirements.

3.7.1 Inch Series. Bits shall be in accordance with Table 2, Torsional Loads, Inch Series. When the shank size is smaller than the tip size, the bit shall meet the torsional requirements of the shank dimension. When torsional testing is done, torsional testing shall be in accordance with ANSI B18.3.2M, paragraphs 3.3.2 Torsional Strength, 3.3.2.1 Torsional Strength Test, and 3.3.2.2 Torsional Shear Test.

Table 2 Torsional Loads (Inch-Pounds)

Nominal Key Size (Inches)	Torsional Load (Inch-Pounds)
0.028	0.94
0.035	1.80
0.050	5.25
1/16	10.25
5/64	20
3/32	35
7/64	55
1/8	82
9/64	118
5/32	160
3/16	278
7/32	440
1/4	665
5/16	1275
3/8	2200
7/16	3500
1/2	5200
9/16	6500
5/8	9000
3/4	15500
7/8	24600
1	28800

3.7.2 Metric Series. Bits shall be in accordance with ANSI B18.3.2, Table 2, Torsional Moments for Torsional strength Test requirements. When the shank size is smaller than the tip size, the bit shall meet the torsional requirements of the shank dimension. When torsional testing is done, torsional testing shall be in accordance with ANSI B18.3.2M paragraphs 3.3.2 Torsional Strength, 3.3.2.1 Torsional Strength Test, and 3.3.2.2 Torsional Shear Test.

3.8 Tip Toughness Test. When a tip toughness test is performed, the torsional load will be increased until inch series bits with 1/4 inch tip and smaller fail or metric series bits 8mm or smaller fail. The

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failure shall occur as a clean relatively square, shear fracture. Any splintering or brittle type failures shall be cause for rejection.

3.9 Sets. The components of sets shall conform to the specified requirements for the individual items.

3.10 Identification marking. The hexagonal screwdriver bits shall be marked with the manufacturer's name or identifying symbol, and the state or country of manufacture, unless otherwise specified. All identification markings shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.11 Workmanship. Details of workmanship shall be in accordance with the best commercial practice. Coatings and finishes shall be shall be smooth, adherent, continuous, and not stained or discolored. External surfaces shall be free from tool and gouge marks, nicks, or other surface imperfections. The item shall be free from manufacturing workmanship defects (e.g., sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety or appearance.

4. REGULATORY REQUIREMENTS.

4.1 Recovered Materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pounds units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5. PRODUCT CONFORMANCE.

5.1 Product Conformance. The product provided shall meet the salient characteristics of the Commercial Item Description, conform to the producer's own drawing, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marked. The government reserves the right to require proof of such conformance.

5.2 Responsibility for Inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any of the inspections deemed necessary to assure the item conforms to the specified requirements.

6. PACKAGING. Preservation, packaging and marking shall be as specified in the contract or order.

7. NOTES. This section contains information of a general or explanatory nature that is not mandatory.

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7.1 Source of Documents.

Federal Acquisition Regulation (FAR): Government Printing Office. Superintendent of Documents, Washington, DC 20402-9371.

American Society for Testing Materials (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

American Society of Mechanical Engineers (ASME),Manuel Gutierrez, Managing Director, 3 Park Avenue, New York, NY 10016.

GSA Specification Section (3FP-E): Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC 20407.

7.2 National Stock Numbers (NSN's). A list of NSN's assigned that correspond to this CID. The list may not be indicative of all possible NSN's associated with the CID.

NSN	Shank Width (Inch)	Bit Size (Inch)	NSN	Shank Width (Inch)	Bit Size (Inch)
5120-00-289-9454	1/4	1/16	5120-00-640-6728	5/16	5/16
5120-00-289-9453	1/4	5/64	5120-00-204-0985	5/16	3/8
5120-00-640-6725	1/4	3/32	5120-00-678-5046	7/16	3/16
5120-00-921-0574	1/4	7/64	5120-00-678-5047	7/16	7/32
5120-00-640-6733	1/4	1/8	5120-00-678-5048	7/16	1/4
5120-00-177-6832	1/4	9/64	5120-00-678-5049	7/16	5/16
5120-00-640-6732	1/4	5/32	5120-00-678-5050	7/16	3/8
5120-00-640-6731	1/4	3/16	5120-00-678-5051	7/16	7/16
5120-00-640-6730	1/4	7/32	5120-00-678-5052	7/16	1/2
5120-00-640-6729	1/4	1/4	5120-00-678-5053	7/16	9/16
5120-00-247-1281	1/4	5/16	5120-00-678-5054	7/16	5/8
Retaining Rings Included					
5120-00-058-5158	1/4	0.050	5120-01-104-3145	5/16	9/64
5120-01-104-3140	1/4	1/16	5120-01-104-3146	5/16	5/32
5120-01-104-3142	1/4	5/16	5120-01-104-3135	5/16	3/16
5120-01-104-3143	1/4	3/32			

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7.3 Ordering data. Purchasers should select the preferred options permitted herein and shall include the following information in procurement documents.

a. Title, number, and date of this Commercial Item Description.
b. The appropriate NSN, or
c. The type, shank size, and point size of each screwdriver bit.
d. Nominal bit length (optional).
e. Retention groove shape (optional).
f. Bit hardness on the HRC scale (optional).
g. Retainer "O" ring shank design (optional).
h. Bit with knurled or serrated spinner on shank end (optional).
i. Specify level of preservation, packaging, and packing.

7.4 Key words. Bit, Screwdriver, Male Shank, Hexagonal Tip

MILITARY INTERESTS:
NONE: DoD has no registered interest in revisions and amendments to this Commercial Item Description until further notice.

Preparing Activity:
GSA-FSS

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